LAB 3: MQTT & ESP32

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I. Introduction

THIS laboratory approaches the connection between data acquisition on an ESP32 and message distribution using MQTT. The guide organizes experiments in connectivity, telemetry, control and serial integration [1].

A. Theoretical Framework

MQTT uses the publisher–subscriber model: a client publishes to a topic, and the server distributes messages according to subscriptions. MQTT 3.1.1 requires an ordered, bidirectional transport, like TCP. Its QoS level 0 offers delivery at most once; it does not guarantee that every message will reach the receiver [2]. Therefore, simply observing a connection is not enough to quantify message loss or receipt of sensor readings.

In the Arduino-ESP32 API, `analogRead` returns a raw, uncalibrated ADC code, while `analogReadMilliVolts` provides a calibrated result in millivolts [3]. An ADC code obtained from a divider with an LDR does not directly equate to illuminance in lux. The relation with illumination requires characterizing the sensor, circuit, and reference conditions; this conversion is not performed here.

PubSubClient 2.8 selects MQTT 3.1.1 by default. Its overload `publish(topic,payload)` constructs a message without retention and without QoS bits 1/2; `loop()` handles client communication [4]. These properties describe the library and allow interpretation of the CP2 sketch; they do not replace a receiver capture.

B. State of the Art

MQTT is commonly studied as a lightweight application-layer protocol for resource-constrained IoT devices. Al-Fuqaha *et al.* identify publish–subscribe protocols, including MQTT, as an important part of the IoT protocol stack and relate protocol selection to device constraints, network conditions, and application requirements [5]. This perspective is relevant to the present experiment because the ESP32 acts as a constrained sensing client while the broker separates the publisher from the monitoring application.

Several studies compare MQTT with alternatives rather than evaluating it in isolation. Thangavel *et al.* evaluated MQTT and CoAP through a common middleware and showed that their behavior depends on the communication pattern and transport assumptions [6]. Similarly, the experimental study by Georgiou and Karapantelakis examined MQTT performance in IoT-to-cloud communication and reported that the selected QoS level affects latency and application behavior [7]. These results support documenting the protocol parameters instead of treating a successful connection as a complete performance evaluation.

For low-power wireless sensor networks, Hunkeler, Truong, and Stanford-Clark proposed MQTT-S as an adaptation of the MQTT model to networks where conventional TCP connectivity is not always available [8]. More recent work by Palmese, Redondi, and Cesana proposed adaptive QoS control for MQTT-S and evaluated its effect on delay and packet errors [9]. In contrast, this laboratory uses standard MQTT over Wi-Fi/TCP and QoS 0 through PubSubClient; therefore, the experiment demonstrates basic telemetry but does not claim the reliability or energy behavior of MQTT-S.

Recent measurements also investigate alternatives to TCP. Pacheco and Santos characterized delay and energy consumption for MQTT over TCP and QUIC using commercial off-the-shelf devices [10]. Their results reinforce that latency and energy depend on the complete transport stack and hardware, not only on the MQTT message format. The present work is consequently positioned as an implementation and functional validation of an ESP32–MQTT telemetry path. It provides evidence of acquisition and delivery, but it does not estimate lux, compare protocols statistically, or measure energy consumption.

II. Methodology

A. Components and CP2 configuration

The CP2 implementation uses an ESP32 development board, an LDR, a fixed resistor, a breadboard, jumper wires, and a USB cable. The LDR and the fixed resistor form a voltage divider. The divider output is connected to GPIO35, which is used as an ADC input by the ESP32. The exact resistance value and the orientation of the divider must match the physical circuit demonstrated during the checkpoint; changing either one changes the direction and scale of the ADC response.

The ESP32 is programmed with the Arduino IDE and the ESP32 board support package. The sketch uses the `WiFi.h` library for network connectivity and PubSubClient 2.8 for MQTT communication. The serial monitor is configured for 115200 baud. Network credentials are stored in `secrets_local.h`, which is intentionally excluded from the repository. The broker address is supplied through `mqtt_broker` and the broker listens on the standard non-TLS MQTT port, 1883. The client publishes the raw ADC value as a text payload to the `lab/ldr` topic.

The program first waits until the ESP32 is associated with the configured Wi-Fi network. It then configures the broker address and port. During each iteration, `analogRead(35)` acquires the raw ADC code, the value is printed to the serial monitor, and the MQTT connection is established if necessary. After `mqtt.loop()` services the client, the value is published and the program waits 1000 ms before the next

iteration. The 1000 ms delay is a minimum application delay rather than an exact sampling period because ADC acquisition, Wi-Fi processing, and MQTT operations add execution time.

B. Hardware and communication sequence

The complete data path is shown in Fig. 1. The sensor circuit produces an analog voltage, the ESP32 samples it, and the MQTT broker forwards the resulting payload to a subscribed client. Because `analogRead` returns an ADC code rather than a calibrated illuminance value, the reported payload is intentionally described as a raw LDR reading.

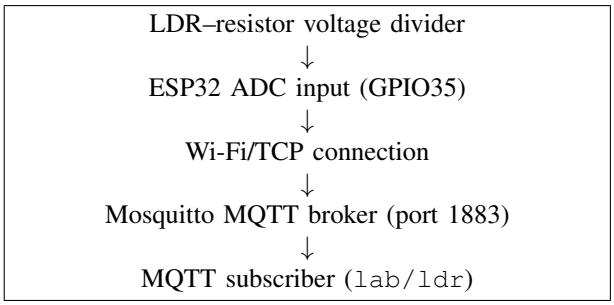

Figure 1. CP2 block diagram for LDR acquisition and MQTT telemetry.

Fig. 2 summarizes the execution sequence. A failed MQTT connection is reported through the serial monitor and causes the current iteration to return before the next attempt.

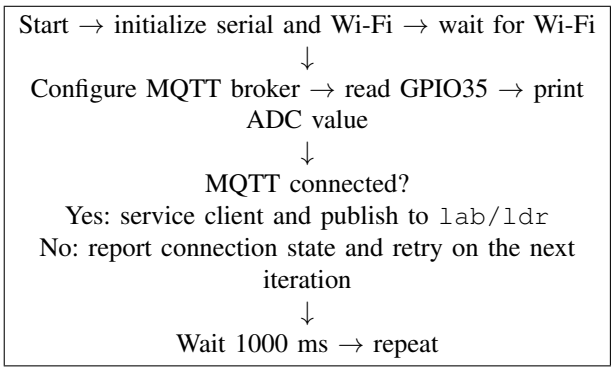

Figure 2. CP2 program flow for acquisition and MQTT publication.

III. Development

The purpose of this section is to answer both the checkpoint and report questions of the lab, as well as discuss any relevant results obtained during the experiences.

A. CHECKPOINT 1: Previous Installations

1) **Arduino Installation and ESP32 Setup**
First, we downloaded and installed the Arduino IDE for our computer's operating system, which in this case was Windows 11, as shown in Figure 3.
Once installed and opened, we needed to add support for the ESP32 board. To do this, we went to File > Preferences. There, we added the URL https://dl.espressif.com/dl/package_esp32_index.json in the "Additional Boards Manager URLs" field and clicked "OK", as shown in Figure 4.
Then, we needed to install the board support package for the ESP32. To do this, we went to Tools > Board > Boards Manager. There, we searched for "esp32" and installed the package provided by Espressif Systems, as shown in Figure 5.

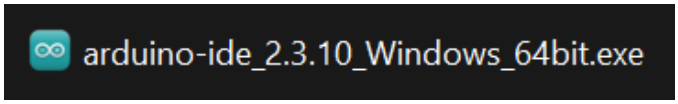

Figure 3. Arduino IDE installation for Windows 11.

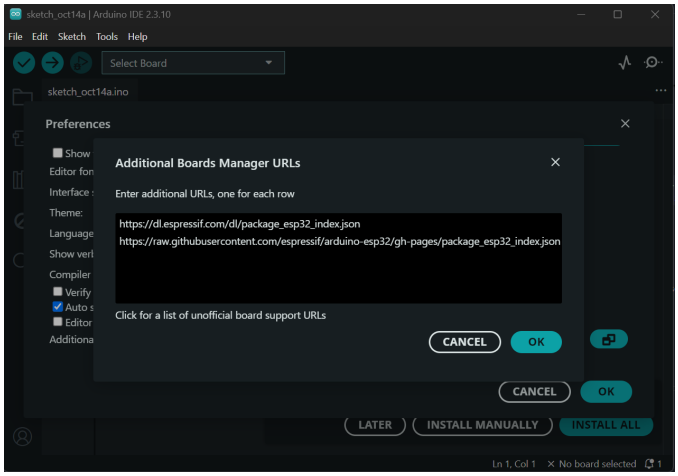

Figure 4. Arduino IDE preferences with the ESP32 board manager URL added.

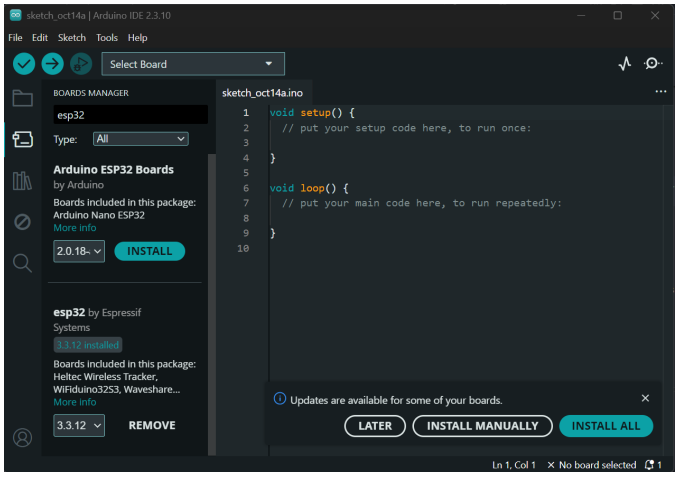

Figure 5. Installation of the ESP32 board support package by Espressif Systems.

To finish the initial setup, we went to Select Board > Select Other Board and Port. There, we searched for "NodeMCU-32S" in the board selection section and selected "COM5 Serial Port" in the ports section, as shown in Figure 6.

2) **Test Code Upload**
We tested a very simple code and actually ran into a

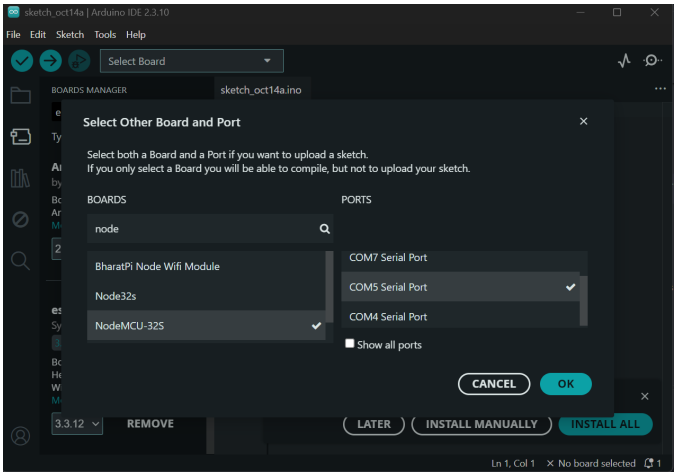

Figure 6. Initial selection of the NodeMCU-32S board and COM5 serial port in Arduino IDE.

small problem during the upload process. As shown in Figure 7, there was an issue with the selected serial port. We solved this by changing “COM5 Serial Port” to “COM3 Serial Port (USB)”, which corresponded to the connected ESP32, as shown in Figure 8. After fixing this issue, we successfully uploaded and ran the test code on the ESP32, as shown in Figures 9 and 10.

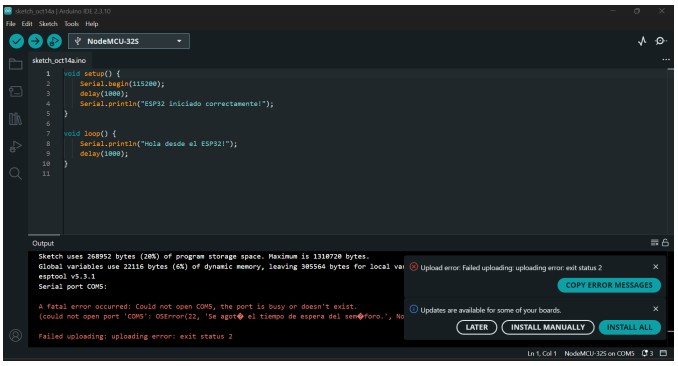

Figure 7. Error encountered during the initial code upload due to the incorrect serial port selection.

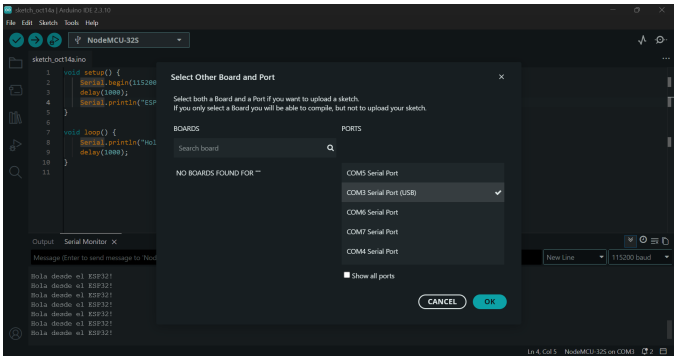

Figure 8. Correction of the serial port selection from COM5 to COM3 (USB).

3) Mosquitto Installation

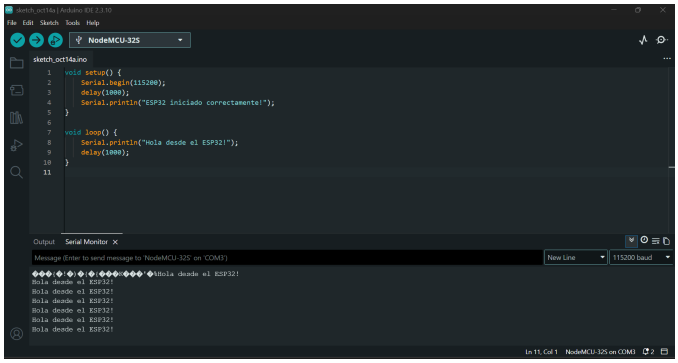

Figure 9. Successful compilation and upload of the test code to the ESP32.

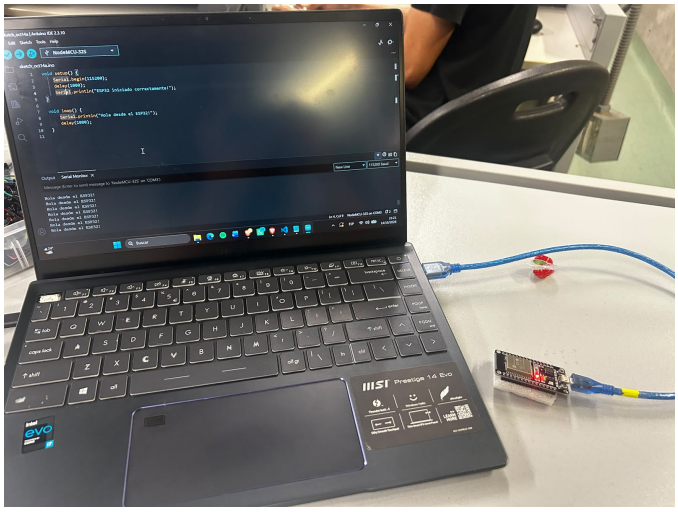

Figure 10. Arduino IDE Serial Monitor displaying the messages generated by the ESP32 test code.

First, we downloaded the Mosquitto executable installer for Windows, as shown in Figure 11. As instructed, Mosquitto was installed outside the Program Files directory. We also needed to add its installation path to the system environment variables in order to execute Mosquitto commands directly from the terminal. We then confirmed that Mosquitto could be executed successfully, as shown in Figure 12.

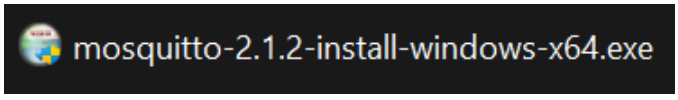

Figure 11. Download of the Mosquitto installer for Windows from its official website.

Then, we modified the Mosquitto configuration file, as shown in Figure 13, and successfully ran the broker using the updated configuration, as shown in Figure 14. Finally, we tested MQTT communication by subscribing to the topic “test/topic” and publishing a test message to the same topic. As shown in Figure 16, we published a message, and as shown in Figure 15, we were able to receive it successfully through the subscribed client.

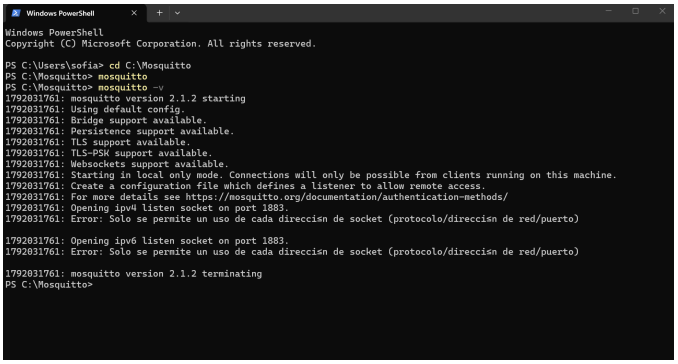

Figure 12. Verification of the Mosquitto installation through the Windows terminal.

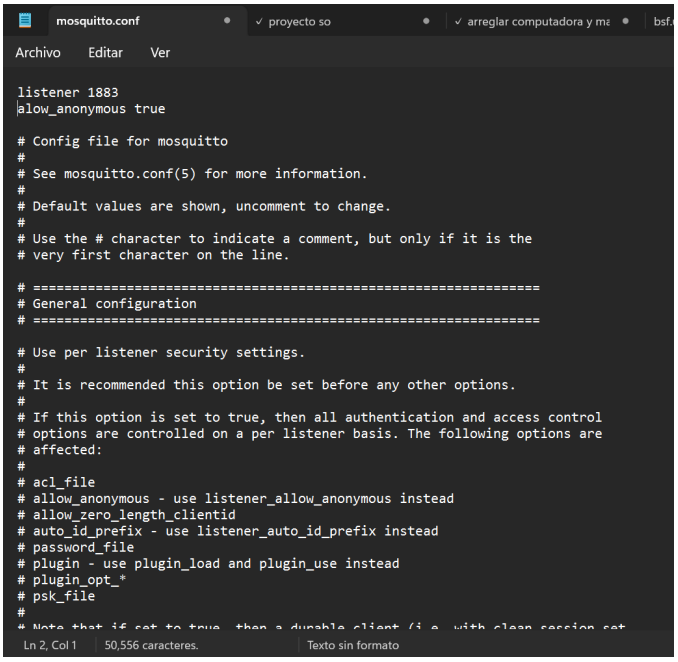

Figure 13. Modification of the Mosquitto configuration file to enable connections on port 1883 and allow anonymous clients.

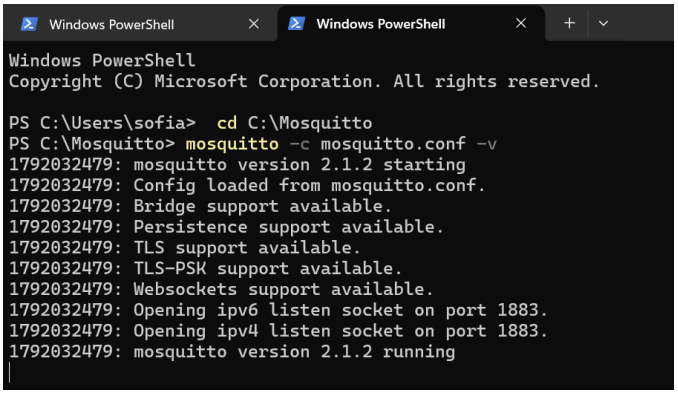

Figure 14. Successful execution of the Mosquitto MQTT broker using the updated configuration file.

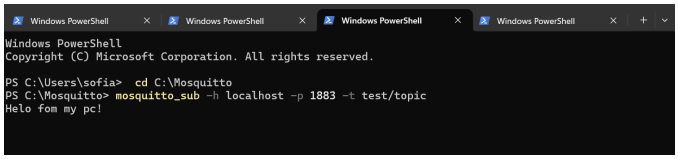

Figure 15. MQTT subscriber receiving the test message through the Mosquitto broker.

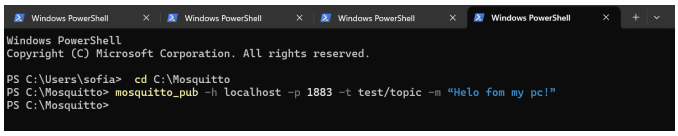

Figure 16. Publication of a test message to the MQTT topic using the Mosquitto command-line client.

B. CHECKPOINT 2: Basic Implementation with Reading

The second checkpoint implemented a complete telemetry path from an LDR connected to the ESP32 ADC to an MQTT subscriber on another computer. The circuit was assembled as a voltage divider, and its output was connected to GPIO35.

1) Implementation Procedure: The implementation followed these steps:

1) The LDR voltage-divider output was connected to GPIO35, while the circuit was supplied from the ESP32-compatible voltage rail and ground. The fixed-resistor value and the physical orientation were recorded from the assembled circuit because they determine the ADC response.
2) The ESP32 board support package and the PubSubClient library were enabled in Arduino IDE. The board and the serial port corresponding to the connected ESP32 were selected.
3) The broker was configured on port 1883 and the subscriber was configured to listen to `lab/ldr`.
4) The sketch was compiled and uploaded to the ESP32. The serial monitor was opened at 115200 baud to verify Wi-Fi association, display the assigned IP address, and observe the ADC readings.
5) A Mosquitto subscriber was started before or during the test. The ESP32 then read the ADC, connected to the broker when necessary, and published each reading as a text payload.
6) The illumination over the LDR was changed while the serial output and the MQTT subscriber were observed. This test verifies that the acquisition path and the publication path respond to the same physical stimulus.

Figure 17 shows the ESP32 program used for the implementation. The code integrates the analog reading of the LDR with the Wi-Fi and MQTT communication functions. In this way, the ESP32 is responsible for both acquiring the sensor data and publishing it to the broker, without requiring an additional microcontroller.

The physical implementation is shown in Figures 18 and 19. The LDR and fixed resistor form a voltage divider, allowing variations in light intensity to be represented as changes in voltage at the ESP32 analog input. This is necessary because

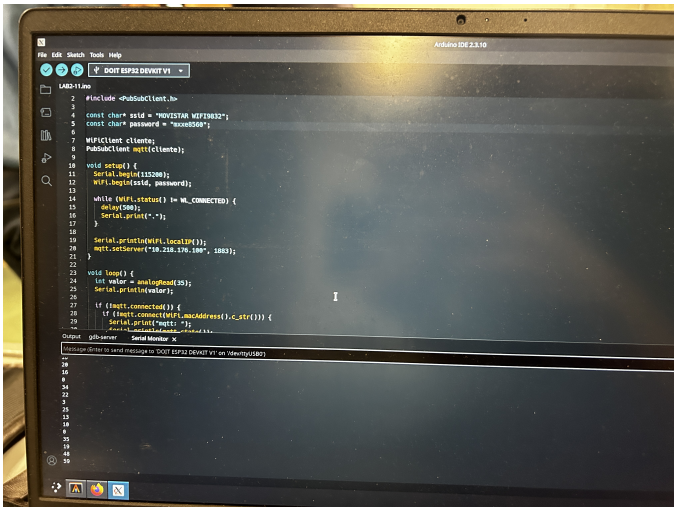

Figure 17. ESP32 program for LDR data acquisition and MQTT communication in Arduino IDE.

the ESP32 measures voltage rather than light intensity directly. The breadboard was used to assemble the circuit and connect the sensor to the development board.

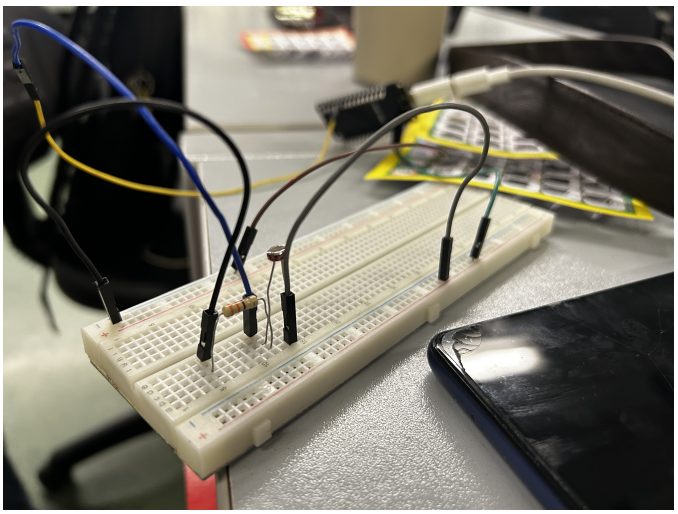

Figure 18. LDR voltage-divider circuit assembled on a breadboard for analog light sensing.

2) Checkpoint Validation and Technical Discussion: The implementation was functionally validated through three main observations: the ESP32 acquired changing ADC readings, the broker received the published messages, and the MQTT subscriber displayed the transmitted values on the receiving computer.

As shown in Figure 20, the publisher-side terminal displayed a continuous sequence of LDR readings generated by the ESP32. This shows that the sensor was being read periodically, while the MQTT subscriber shown in Figure 21 confirms that these readings were also received on the other computer. The publication interval was approximately one second, although the actual time between messages also depends on the ADC, Wi-Fi, and MQTT operations.

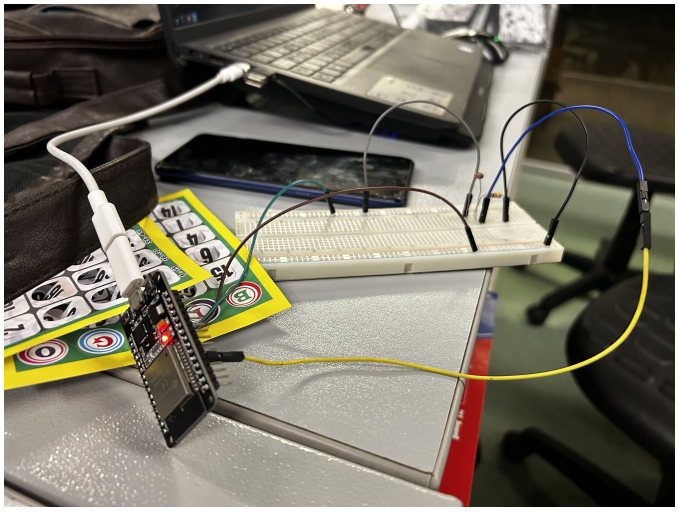

Figure 19. Physical implementation of the LDR sensing circuit connected to the ESP32 development board.

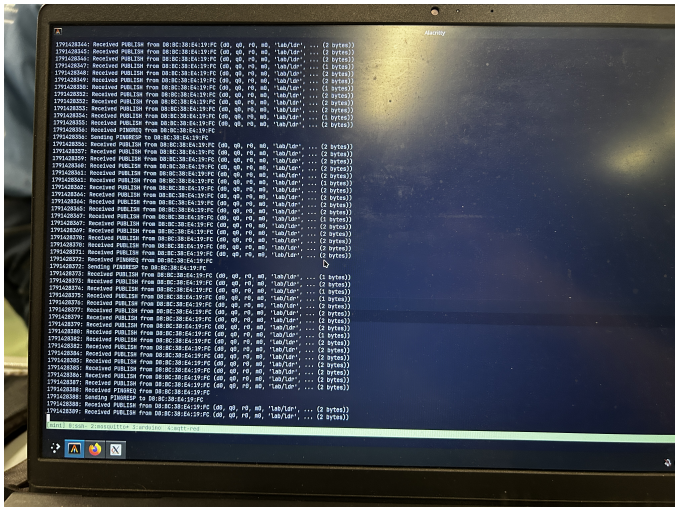

Figure 20. Publisher-side terminal displaying the periodic LDR readings generated by the ESP32.

Figure 21 shows the MQTT subscriber receiving messages under the topic `lab/ldr`. The displayed values changed as the LDR was covered and uncovered, showing that variations in the physical input were reflected in the received data. This is important because it verifies not only that the devices were connected, but also that the information acquired by the sensor reached the other computer through the broker.

The complete demonstration can also be observed in the following video: Checkpoint 2 demonstration video.

In the video, the two-computer communication is demonstrated while the illumination reaching the LDR is manually modified. By covering and uncovering the sensor, the received values change accordingly, providing visual evidence that the system responds to changes in the environment and communicates the updated information remotely. This confirms the basic functionality of the MQTT publisher-subscriber architecture. The readings from the LDR respond to changes in illumination. The implementation uses the default publication

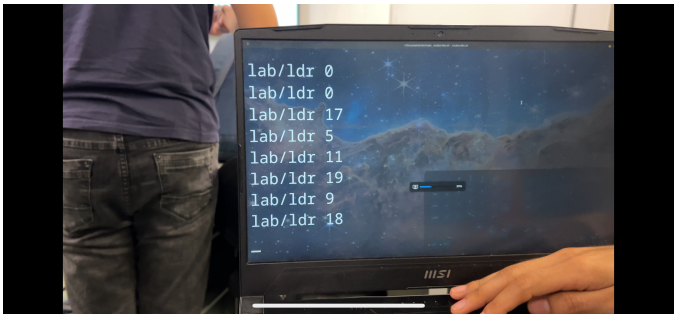

Figure 21. MQTT subscriber displaying changing LDR readings received under the lab/ldr topic.

behavior of PubSubClient, MQTT 3.1.1 with QoS 0. Consequently, receiving the messages demonstrates successful communication during the test, but does not guarantee that every published message was delivered or provide a measurement of packet loss.

3) Block Diagram and Flowchart: The system architecture and the program execution sequence are represented in Figures 22 and 23, respectively.

The block diagram identifies the main components involved in data acquisition and transmission. The LDR provides the analog input to the ESP32, which processes the reading and publishes it through Wi-Fi to the Mosquitto broker. The broker then forwards the message to the subscribed client on the second computer.

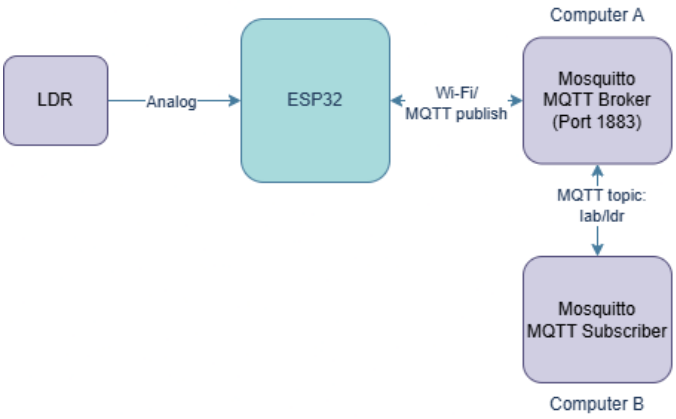

Figure 22. Block diagram of the LDR sensing and MQTT communication system using an ESP32.

The flowchart describes the internal execution of the ESP32 program, including initialization, sensor acquisition, MQTT connection verification, and periodic publication. The connection decision is relevant because the ESP32 must establish communication with the broker before it can successfully publish the acquired data.

C. CHECKPOINT 3: Smart Ventilation and Light Control with MQTT

In this checkpoint, we implemented an automated system that responds to changes in ambient light using an ESP32,

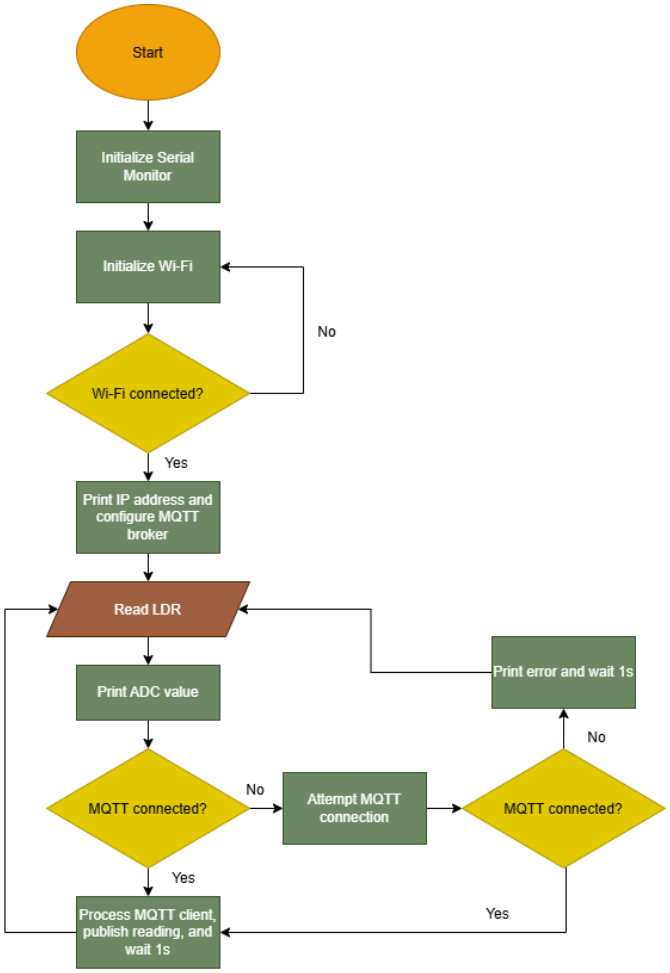

Figure 23. Flowchart of the ESP32 program for periodic LDR readings and MQTT publication.

an LDR, and a potentiometer. The LDR measures the light level, while the potentiometer establishes an adjustable threshold. Depending on the comparison between these values, the system controls a servomotor, a buzzer, and LED indicators. Additionally, MQTT communication allows the servomotor to be controlled remotely through manual commands.

1) Physical Implementation: The system was assembled using an ESP32, an LDR with a fixed resistor, a potentiometer, an SG90 servomotor, a buzzer, three LEDs (red, green, and blue), resistors, a breadboard, and jumper wires. The readings from the LDR and the potentiometer are compared to determine whether the measured light level exceeds the threshold selected by the user. When the threshold is exceeded, the servomotor moves 90° to simulate closing a blind, the red LED turns on, and the buzzer should activate as an audible warning. Otherwise, the servomotor returns to its open position 0°, the green LED turns on, and the buzzer remains off. The blue LED indicates the connection to the MQTT broker.

As shown in Figure 24, the red LED is illuminated while the blue LED remains on. This is consistent with the system entering its alert state when the measured light exceeds the selected threshold. The red LED indicates that the automatic control should close the blind, while the blue LED represents

the MQTT connection status.

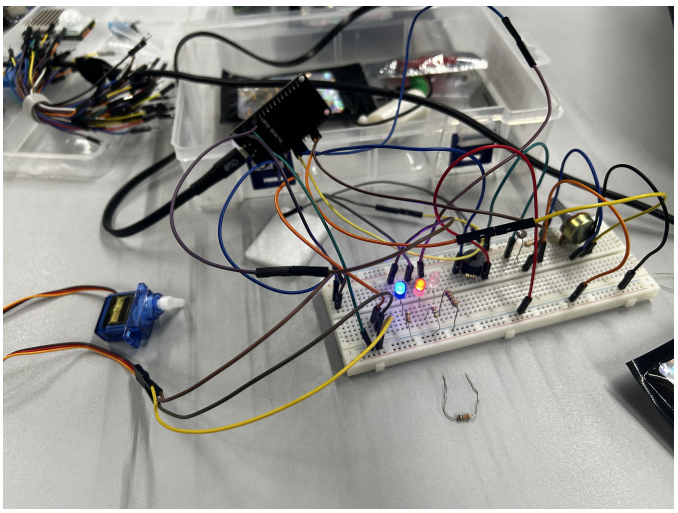

Figure 24. Physical implementation of the automatic light control system with the red and blue LEDs activated.

In Figure 25, the green LED is illuminated instead of the red LED. This corresponds to the normal operating state, in which the measured light does not exceed the configured threshold and the blind should remain open. Comparing both photographs shows how the LED indicators distinguish between the two operating conditions, while the blue LED remains illuminated in both cases.

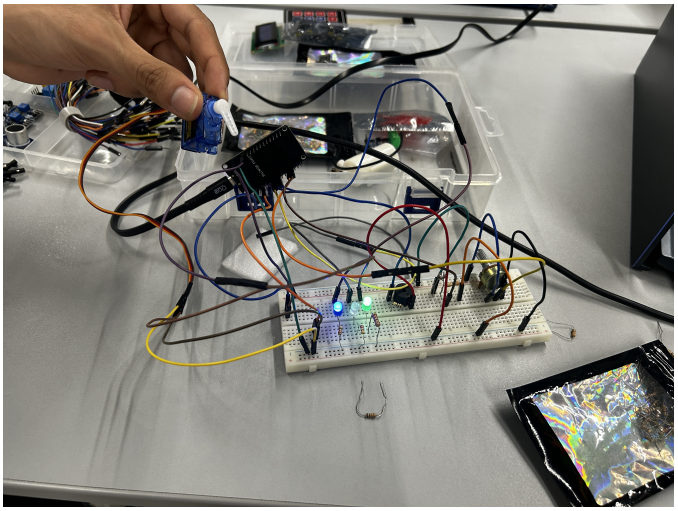

Figure 25. Physical implementation showing the green and blue LEDs activated during normal operation.

2) Automatic Control: To test the automatic mode, we modified the threshold using the potentiometer, covered the LDR to change the light input and observed how the system responded to the LDR reading.

The automatic operation can be observed in the following demonstrations: Automatic control demonstration 1 Automatic control demonstration 2

As shown in the demonstrations, the system responds correctly when the potentiometer is adjusted. When the light reading exceeds the selected threshold, the red LED turns on, the servomotor moves to the closed position, and the buzzer is expected to activate. When the light reading is below the threshold, the green LED turns on and the servomotor returns to the open position. This shows that the automatic control works as expected and responds to changes in the selected threshold.

However, we noticed that the buzzer did not always sound immediately when activated. Sometimes it started working after a while, possibly due to a loose connection, although we could not confirm this since we did not have the opportunity to test another buzzer or replace the wires.

3) Manual Control through MQTT: Besides automatic operation, the system also supports remote control through MQTT commands. For this test, commands were manually published from a terminal connected to the MQTT broker, allowing us to change the servomotor position without depending on the current LDR reading.

The manual operation is shown in the following video: Manual MQTT control demonstration

During this demonstration, the command `cerrar` was sent to move the servomotor to the closing position, while `abrir` moved it back to the open position. The command `auto` was then used to return the system to automatic operation, allowing the servomotor to respond again to the comparison between the LDR reading and the potentiometer threshold.

This test is relevant because it demonstrates that the ESP32 can operate as both an MQTT publisher and subscriber. While the sensor measurements and system status can be published to the broker, the ESP32 can also receive commands to modify the behavior of an actuator.

Figures 26 and 27 show the MQTT communication during the demonstration. In Figure 26, the commands `abrir` and `cerrar` were published to the `iluminacion/control` topic to control the servomotor remotely. Figure 27 also shows the Mosquitto broker receiving PUBLISH messages on the `iluminacion/datos` topic, indicating that the ESP32 was transmitting telemetry while the system was operating. Together with the manual control video, these results show that MQTT communication was used for both monitoring and remote control.

Figure 26. Mosquitto terminal showing MQTT control messages used to change the servomotor position and operating mode.

4) Block Diagram and Flowchart: The system architecture is represented in Figure 28. The block diagram identifies the

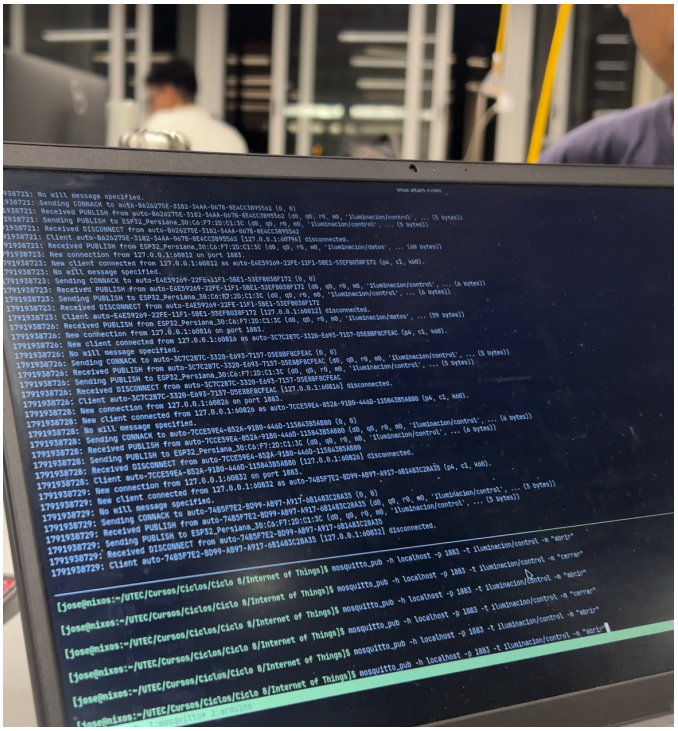

Figure 27. Mosquitto broker logs showing MQTT client connections and message activity during the system demonstration.

sensing components, the ESP32 controller, the actuators, and the MQTT communication path.

The LDR and potentiometer provide analog inputs to the ESP32, which determines the operating condition and controls the servomotor, buzzer, and LEDs. Through Wi-Fi, the ESP32 communicates with the Mosquitto broker to publish telemetry and receive remote commands.

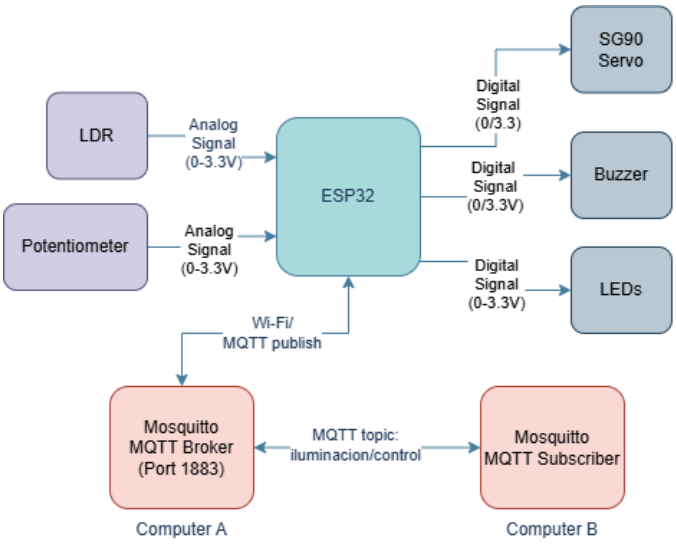

Figure 28. Block diagram of the automatic light control system with local actuators and MQTT communication.

The program execution sequence is shown in Figure 29. The flowchart includes the initialization of the ESP32, acquisition of the analog readings, verification of the MQTT connection, and selection between automatic and manual operation. In automatic mode, the system compares the LDR reading with the threshold and changes the actuator states accordingly. In manual mode, the servomotor responds to the received MQTT commands. This separation allows remote control while maintaining the option to return to autonomous operation.

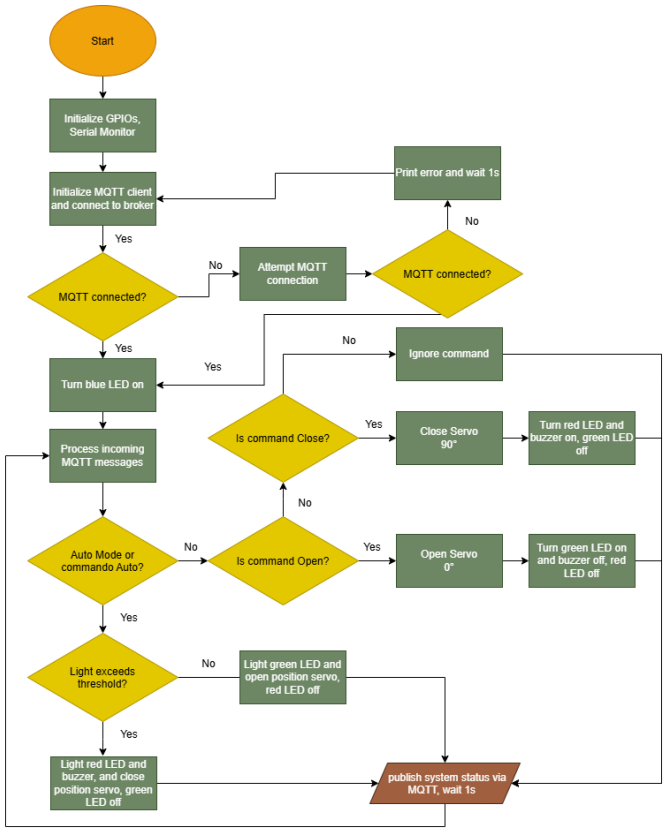

Figure 29. Flowchart of the ESP32 program including automatic threshold-based control and manual MQTT commands.

D. CHECKPOINT 4: Arduino Mega + ESP32 Communication

In this checkpoint, we implemented a small system for simulating microcontroller communication and data transmission via UART and MQTT.

After connecting the Arduino MEGA and ESP32 communication pins, we were able to transmit mock sensor data from the Arduino, receive it successfully on the ESP32, and publish to an MQTT topic, which we subscribed to from a laptop running a Mosquitto client. A voltage divider was required to avoid damaging the ESP, which runs at $3.3V$.

1) Physical Implementation: To enable communication between these two microcontrollers, their serial communication pins must be connected to each other. As their GND reference must also be the same, we connected their ground pins as well.

- Arduino RX1: connected to ESP32 TX2
- Arduino TX1: connected to ESP32 RX2 (with v.divider)
- Arduino GND: connected to ESP32 GND

For the voltage divider, we used three $10k\Omega$ resistors to get $3.33V$ and $1.67V$. We used the first output node, as the ESP32 expects approximately $3.3V$.

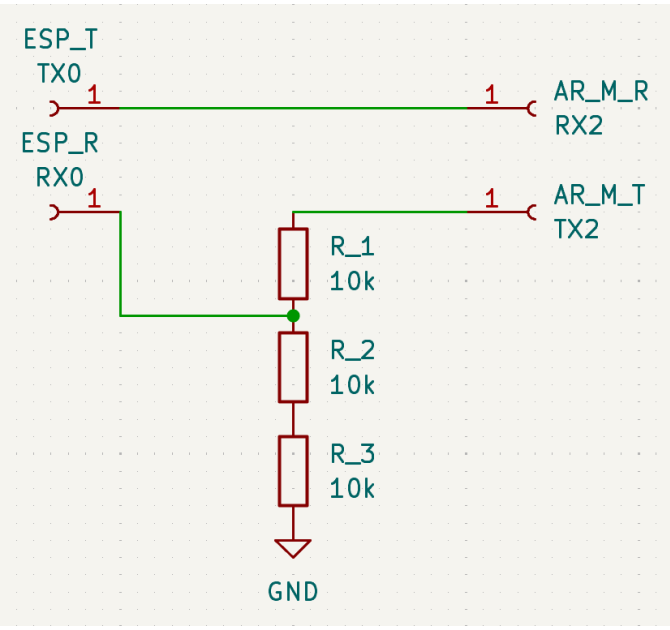

Figure 30. Simple schematic showing necessary connections.

The R and T used to name the pins stand for **receive** and **transmit**, which, despite being apparent, took us a minute to notice (sorry).

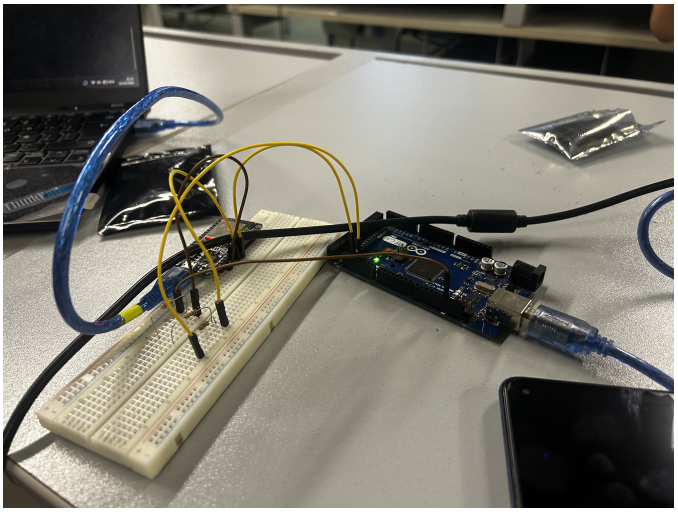

Figure 31. Physical system with a voltage divider to avoid frying the ESP.

2) Arduino Code Upload: The Arduino Mega must mock sensor data (or simple messages) and send them via UART to the ESP32. After physically connecting the pins, we can use *Serial.println()* to send a string to the ESP32. However, as the *Serial* object is usually reserved for the USB connection, we used *Serial1* instead. We do this on a loop with a $1000ms$ delay.

3) ESP32 Code Upload: The ESP32 has a more complex task. It must first connect to a WiFi network, set up an MQTT server, receive messages via UART, and publish them to an MQTT topic.

To make sure we didn't run into any issues with the WiFi network, we set up a mobile hotspot using cellular data. The SSID used was **quaso**.

In the setup function, besides initializing both the *Serial* and *Serial2* objects, we connect to the mobile hotspot, set up an MQTT server on port 1883 and then attempt to connect to the MQTT broker.

```
void setup() {
  Serial.begin(115200);
  Serial2.begin(9600, SERIAL_8N1, RXD2, TXD2);

  WiFi.begin(ssid, password);
  Serial.println(WiFi.localIP());
  mqtt.setServer(mqtt_server, 1883);

  while (!mqtt.connected()) {
    Serial.print("Connecting to MQTT...");

    if (mqtt.connect("esp32cito")) {
      Serial.println("connected!");
    } else {
      Serial.print("failed, state=");
      Serial.println(mqtt.state());
      delay(2000);
    }
  }
}
```

Figure 32. ESP32 code setup

After the setup, the ESP32 reads a string from RX2 and, if valid, publishes it to the MQTT topic. It does this in a loop, while also attempting to reconnect if case the connection with the broker fails.

4) MQTT Monitoring: After flashing both the Arduino and ESP, and succesfully setting up the config file in checkpoint 1, we ran the Mosquitto broker and, on a separate terminal session, subscribed to the MQTT topic with the following commands:

- `./mosquitto -c ./mosquitto.conf -v`
- `./mosquitto_sub h localhost -p 1883 -t hey/guys`

Then, we powered on both microcontrollers, and watched how the laptop received the messages sent by the ESP32.

IV. Conclusion

Conclusiones y recomendaciones experimentales: por documentar.

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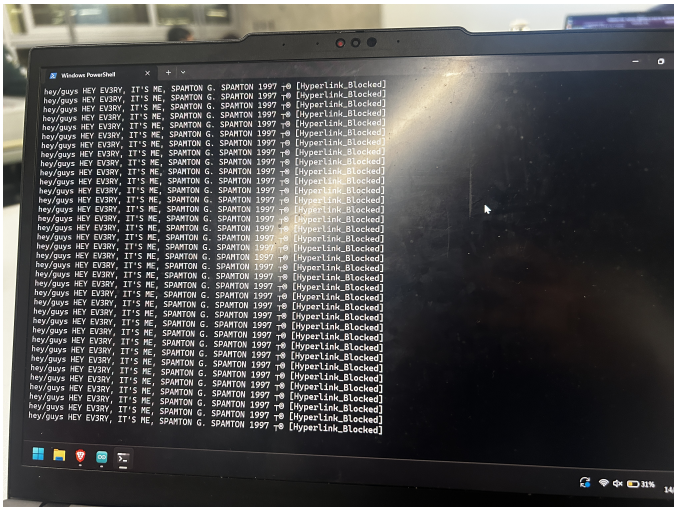

Figure 33. Messages sent by ESP32 shown on the Mosquitto client

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